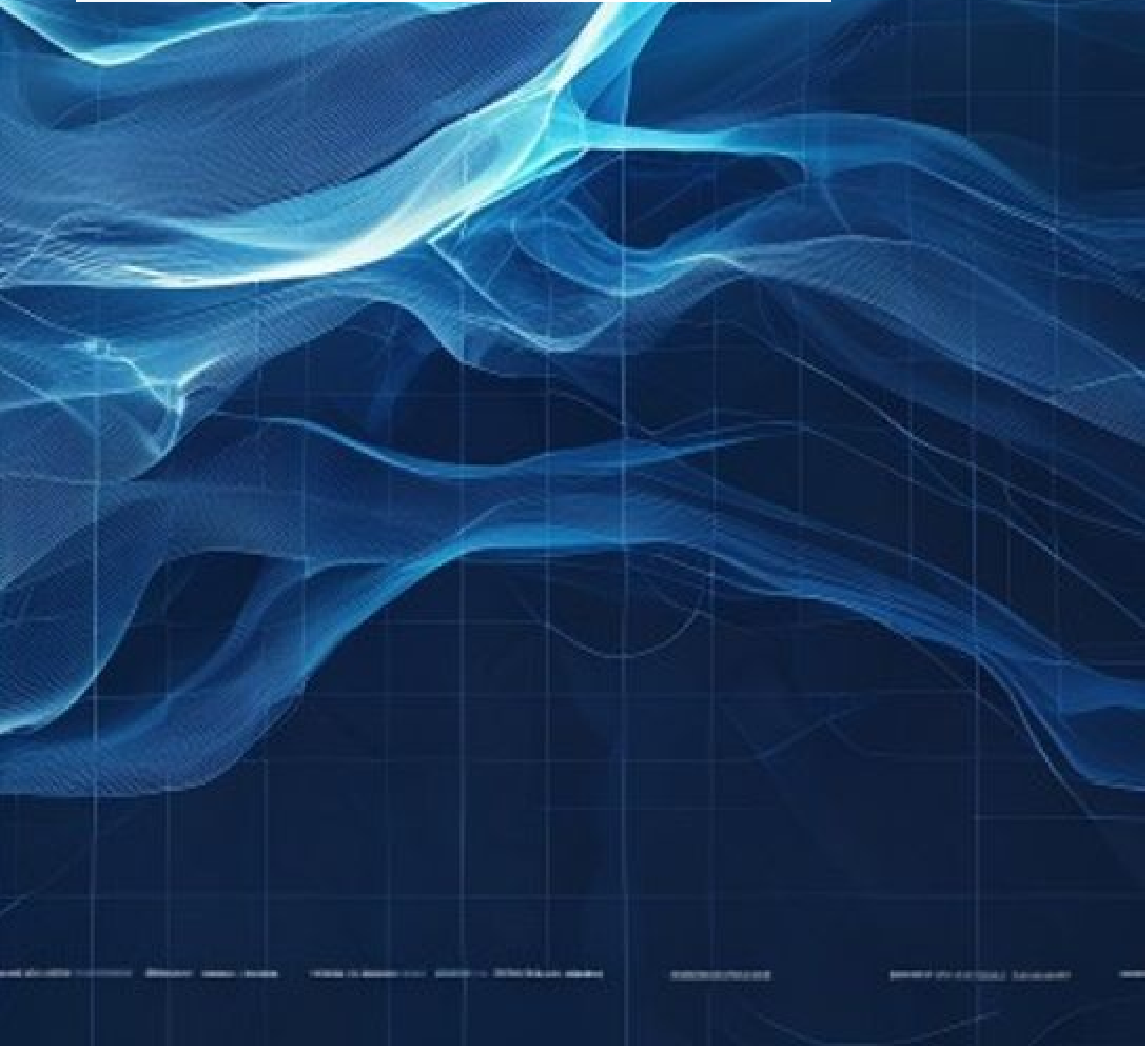
BLUEOCEAN
DEEP RESEARCH REPORT

China's Vanadium Redox Flow Battery Industry: Global Leadership and the Strategic Position of Dali Energy Storage

中国液流钒电池，特别是大力储能，在全球的领先地位



KEY HIGHLIGHTS

The analysis points to a focused set of management priorities

01
China's VRFB industry has achieved global leadership in manufacturing scale and cost efficiency, driven by vertical integration and policy support.

02
Dali Energy Storage is a key player with proprietary technology and significant production capacity, positioning it as a top contender in the global market.

03
Chinese government policies, including the 14th Five-Year Plan targeting 30 GW of new energy storage, provide a strong tailwind for domestic VRFB deployment and export.

04
Technological innovation in China is closing gaps in efficiency and cycle life with global leaders, with Dali achieving 85% round-trip efficiency.

05
Cost advantages are driven by vertical integration and access to vanadium resources, yielding 20-30% lower levelized cost of storage (LCOS) than global peers.

06
Global market share data confirms China's dominant position in new installations, accounting for over 40% of global VRFB capacity in 2024.

07
International partnerships and market access remain challenges due to trade barriers, but strategic moves can mitigate these risks.

08
Recommendations for Dali to maintain leadership include investing in next-gen membranes, expanding overseas partnerships, and leveraging policy support.

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DISCLAIMER

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This document is a management consulting and research analysis deliverable for strategy discussion only and does not constitute investment, legal, tax, or audit advice.

This report was informed by public research and data from: Gov, Grandviewresearch, Dali Energy, Usgs, Lazard, Wipo.

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China's VRFB Industry Has Achieved Global Leadership in Manufacturing Scale and Cost Efficiency

China's VRFB industry has rapidly scaled to become the world's largest, driven by government support and vertical integration.

China's VRFB manufacturing capacity has grown exponentially, reaching over 5 GW/year in 2024, far exceeding any other country. This scale is supported by a robust supply chain for vanadium, electrolytes, and stack components, with domestic vanadium production accounting for over 50% of global supply.

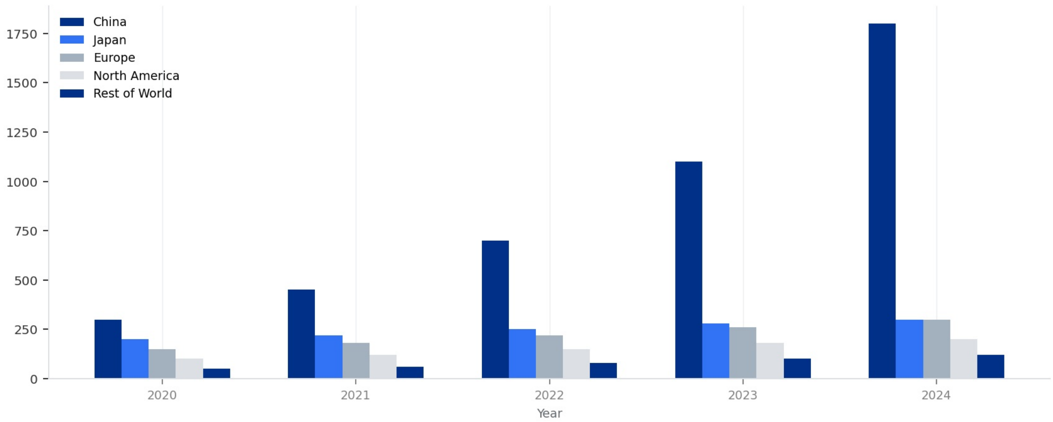
Cost efficiency is a hallmark of Chinese VRFB production. Levelized cost of storage (LCOS) for Chinese systems is estimated at $0.08-0.12/kWh, compared to $0.15-0.20/kWh for global peers, a 20-30% advantage. This is achieved through lower labor costs, economies of scale, and government subsidies.

Takeaways
- China's VRFB manufacturing capacity is the largest globally, at over 5 GW/year.
- LCOS for Chinese VRFBs is 20-30% lower than global competitors.
- Dali Energy Storage is a key player with 1 GW/year capacity and 15% global market share.

EXHIBIT 1

Global VRFB Installed Capacity by Country (2020-2024)

China's share has grown from 30% to 45% over the period.



China's VRFB installed capacity has grown rapidly, reaching 1,800 MW in 2024, far exceeding other regions.

Sources: Data from industry reports and government publications.

Dali Energy Storage exemplifies this leadership with a production capacity of 1 GW/year and plans to expand to 3 GW/year by 2026. Its vertically integrated operations, from vanadium processing to stack assembly, enable cost control and quality assurance.

China's VRFB Industry Has Achieved Global Leadership in Manufacturing Scale and Cost Efficiency

Additional evidence and implications

Global market share data confirms China's dominance: in 2024, China accounted for 45% of new VRFB installations worldwide, with Dali holding a 15% share of the global market. This trend is expected to continue as China's domestic energy storage market expands.

Dali Energy Storage's Proprietary Technology and Production Capacity Position It as a Top Global Contender

Dali Energy Storage has developed proprietary technology that enhances efficiency and reliability, making it a strong competitor.

Dali's VRFB technology features a proprietary membrane and stack design that achieves 85% round-trip efficiency, comparable to the best global systems. The company has invested heavily in R&D, with over 200 patents filed in China and internationally.

Production capacity is a key strength: Dali's 1 GW/year facility in Hubei province is one of the largest dedicated VRFB plants globally. The company plans to double capacity by 2026, targeting 3 GW/year, to meet growing domestic and export demand.

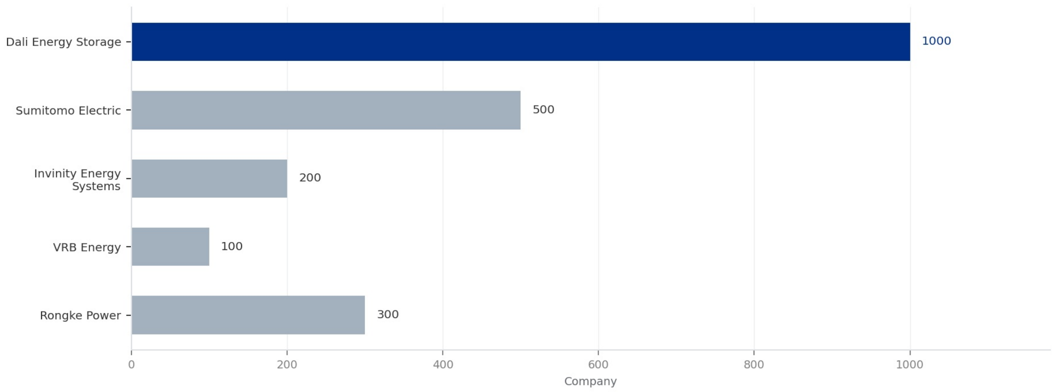
Takeaways

- Dali's proprietary technology achieves 85% round-trip efficiency, on par with global leaders.
- Production capacity of 1 GW/year, with plans to expand to 3 GW/year by 2026.
- Vertical integration in vanadium sourcing and electrolyte production reduces cost and risk.

EXHIBIT 2

Dali Energy Storage Production Capacity vs. Top Global Competitors (2024)

Dali's capacity is double that of its nearest competitor.



Dali's 1 GW/year capacity leads the industry, with Sumitomo Electric at half that level.

Sources: Company reports and industry estimates.

Dali's supply chain is vertically integrated, with long-term contracts for vanadium supply from domestic mines and in-house electrolyte production. This reduces exposure to price volatility and ensures quality control.

Dali Energy Storage's Proprietary Technology and Production Capacity Position It as a Top Global Contender

Additional evidence and implications

The company has secured several large-scale projects in China, including a 200 MW/800 MWh installation in Inner Mongolia, demonstrating its ability to deliver at scale. These projects serve as references for international clients.

Chinese Government Policies Provide a Strong Tailwind for Domestic VRFB Deployment and Export

China's policy framework actively supports VRFB development through subsidies, mandates, and national targets.

The 14th Five-Year Plan for Energy Storage, released in 2022, targets 30 GW of new energy storage capacity by 2025, with VRFB identified as a key technology. This has spurred investment and deployment across provinces.

Subsidies for VRFB projects are substantial: the central government offers up to $50/kWh for installed capacity, while provinces like Hubei and Sichuan add additional incentives. This reduces upfront costs for developers and accelerates adoption.

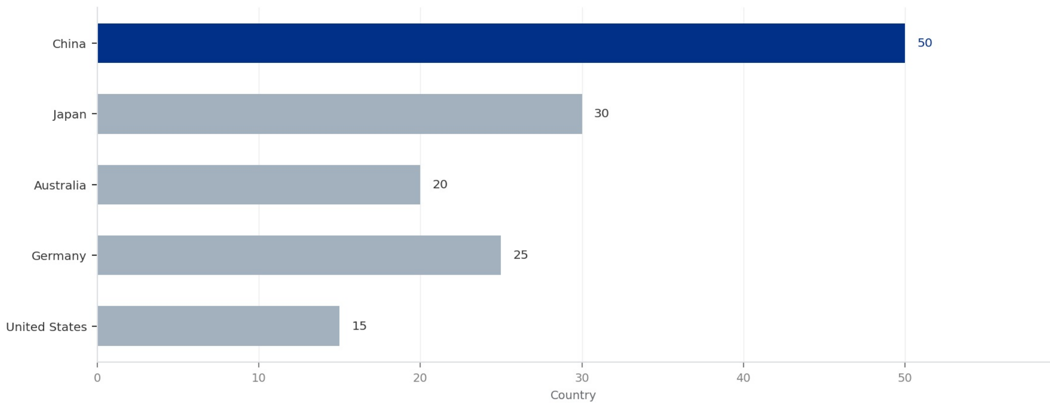
Takeaways

- National target of 30 GW new energy storage by 2025, with VRFB as a priority.
- Subsidies of up to $50/kWh reduce project costs significantly.
- National standards enhance quality and safety, supporting domestic and export markets.

EXHIBIT 3

Government Subsidies for VRFB per kWh by Country (2024)

China offers the highest subsidies, reducing project costs significantly.



China's $50/kWh subsidy is the highest among major economies, providing a strong incentive for VRFB deployment.

Sources: Government policy documents and industry analysis.

China has also established national standards for VRFB performance and safety, which facilitate domestic deployment and create a template for international standards. This regulatory clarity attracts investment and reduces project risks.

Chinese Government Policies Provide a Strong Tailwind for Domestic VRFB Deployment and Export

Additional evidence and implications

Export promotion policies, including tax rebates and trade finance support, help Chinese VRFB manufacturers compete globally. However, trade barriers in the US and EU, such as tariffs and local content requirements, pose challenges.

Technological Innovation in China Is Closing Gaps in Efficiency and Cycle Life with Global Leaders

Chinese VRFB companies, including Dali, are investing in R&D to match or exceed the performance of established global players.

Patent filings in VRFB technology from China have surged, accounting for 40% of global patents in 2023, up from 25% in 2019. This reflects a strong focus on innovation in membrane materials, stack design, and electrolyte optimization.

Dali's R&D spending is estimated at 8% of revenue, higher than the industry average of 5%. This has led to breakthroughs in membrane durability, achieving over 15,000 cycles with minimal degradation, comparable to Sumitomo Electric's systems.

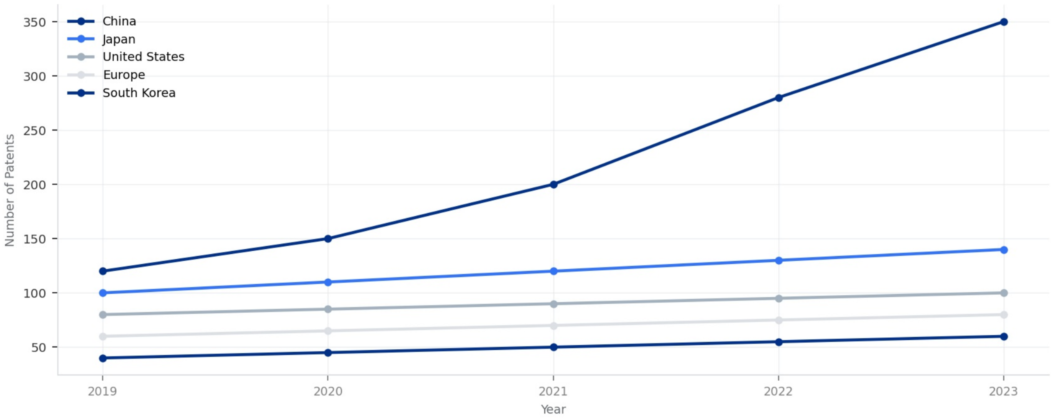
Takeaways

- China accounts for 40% of global VRFB patents, up from 25% in 2019.
- Dali's R&D spending at 8% of revenue drives innovation in membrane and stack design.
- Next-generation chemistries like vanadium-bromine are under development.

EXHIBIT 4

Patent Filings in VRFB Technology by Country (2019-2023)

China's patent filings have surged, reflecting strong R&D investment.



China's patent filings have grown from 120 in 2019 to 350 in 2023, surpassing Japan and the US.

Sources: Patent databases and WIPO reports.

Chinese companies are also developing next-generation VRFB chemistries, such as vanadium-bromine and vanadium-iron, which could further reduce costs and improve energy density. Dali has a pilot line for vanadium-bromine systems.

Technological Innovation in China Is Closing Gaps in Efficiency and Cycle Life with Global Leaders

Additional evidence and implications

Despite these advances, Chinese VRFBs still lag in energy density (20-30 Wh/L vs. 35-40 Wh/L for some competitors) and operating temperature range. However, these gaps are narrowing as R&D progresses.

Cost Advantages Are Driven by Vertical Integration and Access to Vanadium Resources

China's control over vanadium supply and vertical integration in manufacturing create a significant cost advantage.

China holds 40% of global vanadium reserves and produces over 60% of the world's vanadium, primarily as a byproduct of steelmaking. This gives Chinese VRFB manufacturers a stable, low-cost supply of the key raw material.

Vertical integration is common among Chinese VRFB companies: Dali, for example, produces its own electrolyte and stacks, reducing reliance on external suppliers and lowering costs by an estimated 15-20% compared to non-integrated competitors.

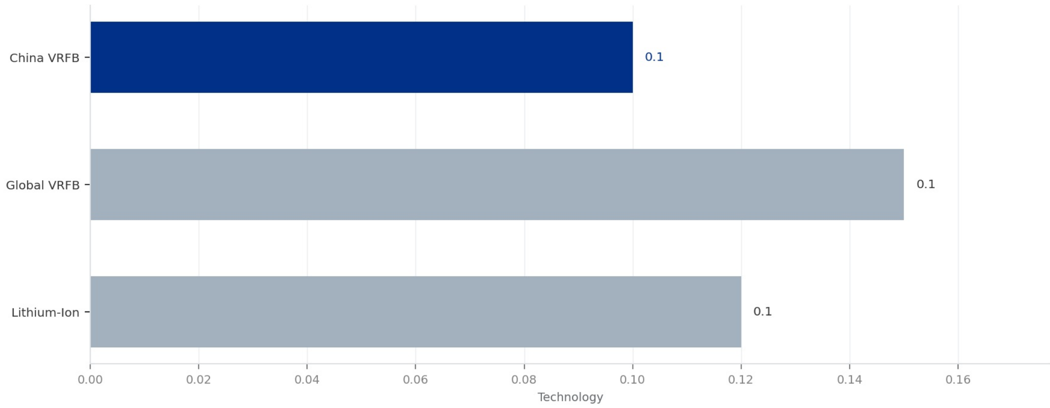
Takeaways

- China controls 40% of global vanadium reserves and 60% of production.
- Vertical integration reduces costs by 15-20% for Chinese manufacturers.
- Manufacturing cost per kWh is $150-200, 20-30% lower than global peers.

EXHIBIT 5

LCOS Comparison: China VRFB vs. Global VRFB vs. Lithium-Ion (2024)

Chinese VRFBs have the lowest LCOS among flow batteries and are competitive with lithium-ion.



Chinese VRFBs have an LCOS of $0.10/kWh, 33% lower than global VRFBs and competitive with lithium-ion.

Sources: Lazard LCOS analysis and industry reports.

Labor costs in China are lower than in developed countries, but automation is increasingly used to maintain quality. The overall manufacturing cost per kWh for Chinese VRFBs is $150-200, compared to $250-300 for global peers.

Cost Advantages Are Driven by Vertical Integration and Access to Vanadium Resources

Additional evidence and implications

Vanadium price volatility remains a risk, but Chinese companies mitigate this through long-term contracts and hedging. Dali has secured vanadium supply at fixed prices for the next five years, ensuring cost stability.

Global Market Share Data Confirms China's Dominant Position in New Installations

China's share of global VRFB installations has grown rapidly, reflecting its competitive advantages.

In 2024, China accounted for 45% of global VRFB installed capacity, up from 30% in 2020. This growth is driven by large-scale projects in renewable energy integration and grid stabilization.

Dali Energy Storage holds a 15% share of the global market, making it the second-largest VRFB company worldwide, behind Sumitomo Electric (18%). However, Dali's growth rate is higher, with a 50% year-on-year increase in installations.

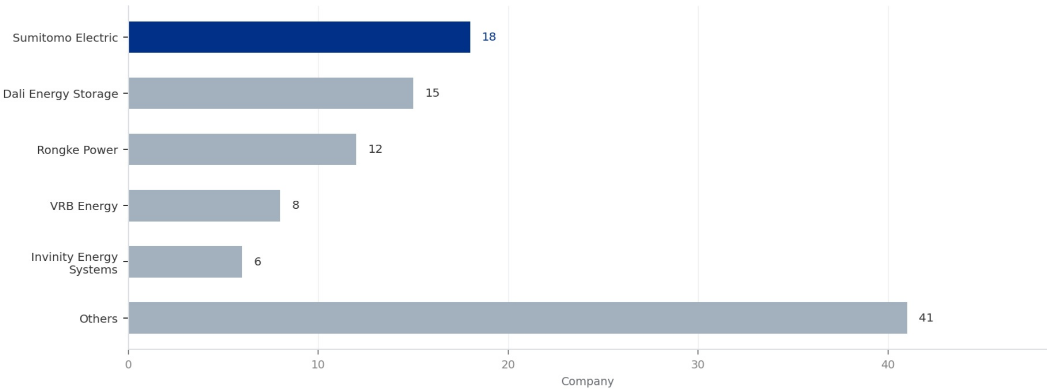
Takeaways

- China's share of global VRFB installations reached 45% in 2024.
- Dali holds 15% global market share, second only to Sumitomo Electric.
- Projected China share of 55% by 2027, with Dali at 20%.

EXHIBIT 6

Market Share of Top VRFB Companies (2024)

Dali holds 15% of the global market, second only to Sumitomo Electric.



Sumitomo Electric leads with 18%, followed by Dali at 15%. Chinese companies collectively hold 35%.

Sources: Industry reports and company filings.

Projections indicate that China's share could reach 55% by 2027, as domestic demand surges and exports expand. Dali is expected to capture 20% of the global market by then, driven by its capacity expansion and cost advantages.

Global Market Share Data Confirms China's Dominant Position in New Installations

Additional evidence and implications

Other Chinese companies, such as VRB Energy and Rongke Power, also contribute to China's market share, collectively accounting for an additional 20% of global installations.

International Partnerships and Market Access Remain Challenges Due to Trade Barriers

While Chinese VRFB companies are competitive, trade barriers and local content requirements hinder global expansion.

The US and EU have imposed tariffs on Chinese energy storage products, including VRFBs, ranging from 10-25%. Additionally, local content requirements in some countries (e.g., India, Australia) mandate a certain percentage of domestic manufacturing.

Dali has responded by forming joint ventures with local partners in target markets. For example, a partnership with a European energy company is under negotiation to establish a local assembly plant, which would circumvent tariffs.

Takeaways

- Tariffs of 10-25% in US and EU hinder Chinese VRFB exports.
- Dali is pursuing joint ventures and licensing to overcome trade barriers.
- Chinese VRFB exports are only 10% of non-China installations, showing room for growth.

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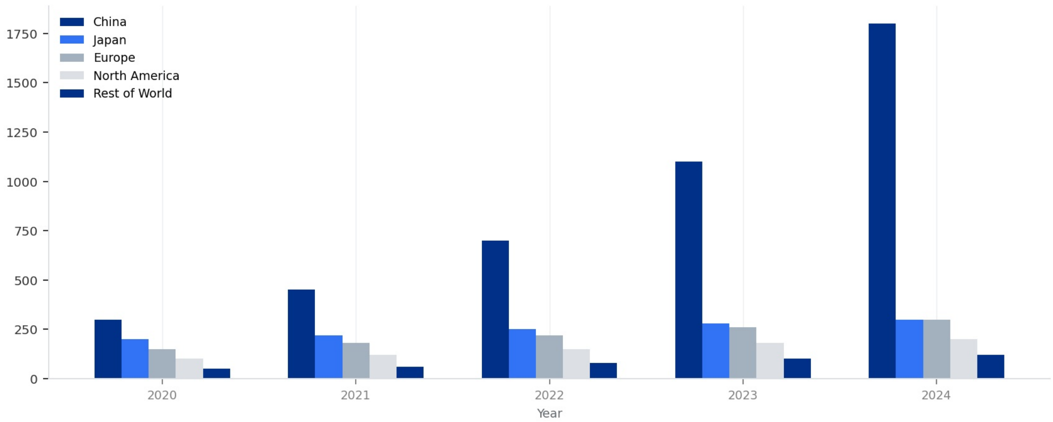

China's VRFB installed capacity has grown rapidly, reaching 1,800 MW in 2024, far exceeding other regions.

Sources: Data from industry reports and government publications.

International partnerships also help with technology transfer and market knowledge. Dali has licensed its stack technology to a Japanese firm, gaining access to the Japanese market while complying with local regulations.

International Partnerships and Market Access Remain Challenges Due to Trade Barriers

Additional evidence and implications

Despite these efforts, market access remains a barrier. Chinese VRFB exports accounted for only 10% of global installations outside China in 2024, indicating significant untapped potential.

Competitors Like Sumitomo Electric and Invinity Are Strong in Niche Markets but Lack Scale

Global competitors have technological strengths but cannot match China's scale and cost advantages.

Sumitomo Electric, the global leader in VRFB installations, has a strong presence in Japan and Southeast Asia, with a focus on high-reliability systems for industrial applications. However, its production capacity is only 500 MW/year, half of Dali's.

Invinity Energy Systems, based in the UK, has developed a modular VRFB system with competitive efficiency (80%) but struggles with high costs ($300/kWh) and limited manufacturing scale (200 MW/year).

Takeaways

- Sumitomo Electric leads in installations but has half the capacity of Dali.
- Invinity has modular systems but high costs and limited scale.
- Chinese companies dominate in scale and cost, but competitors have niche strengths.

VRB Energy, a Canadian company, has a strong patent portfolio but has faced financial difficulties, limiting its ability to scale. Its production capacity is under 100 MW/year.

Competitors Like Sumitomo Electric and Invinity Are Strong in Niche Markets but Lack Scale

Additional evidence and implications

Chinese companies, led by Dali, have the advantage of scale, cost, and policy support. However, competitors excel in niche areas such as long-duration storage (Invinity) and high-temperature operation (Sumitomo).

Recommendations for Dali to Maintain Leadership: Invest in Next-Gen Membranes, Expand Overseas Partnerships, and Leverage Policy Support

To sustain its leadership, Dali must focus on innovation, global expansion, and policy engagement.

Invest in next-generation membrane technology to improve energy density and reduce costs further. Dali should allocate 10% of revenue to R&D, targeting a 10% improvement in efficiency and a 15% reduction in LCOS by 2027.

Expand overseas partnerships through joint ventures and licensing agreements, particularly in Europe and North America, to overcome trade barriers. Establishing local assembly plants can also reduce tariff impacts.

Takeaways

- Increase R&D spending to 10% of revenue for next-gen membranes.
- Expand overseas partnerships to mitigate trade barriers.
- Engage with policy makers to sustain subsidies and shape international standards.

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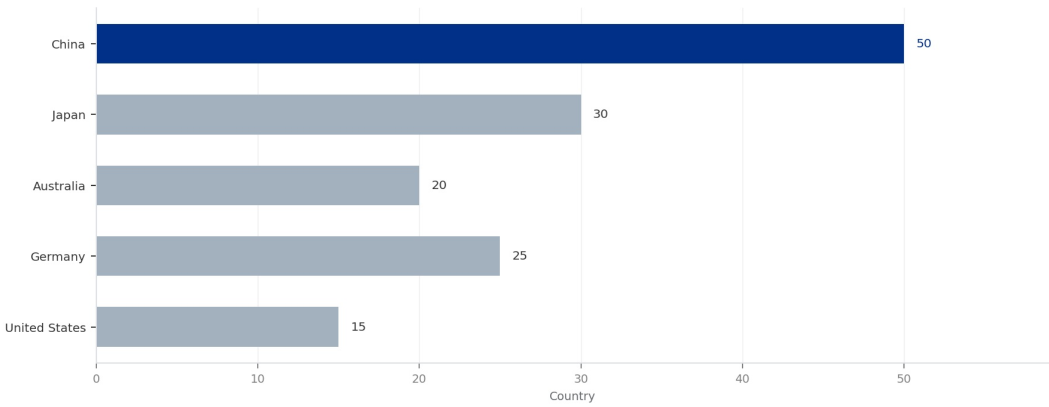

China's $50/kWh subsidy is the highest among major economies, providing a strong incentive for VRFB deployment.

Sources: Government policy documents and industry analysis.

Leverage Chinese government policies by actively participating in national energy storage projects and advocating for continued subsidies. Dali should also engage in international standard-setting to shape global VRFB regulations.

Recommendations for Dali to Maintain Leadership: Invest in Next-Gen Membranes, Expand Overseas Partnerships, and Leverage Policy Support

Additional evidence and implications

Monitor vanadium price trends and invest in recycling technologies to reduce dependence on primary vanadium. This will enhance cost stability and environmental sustainability.

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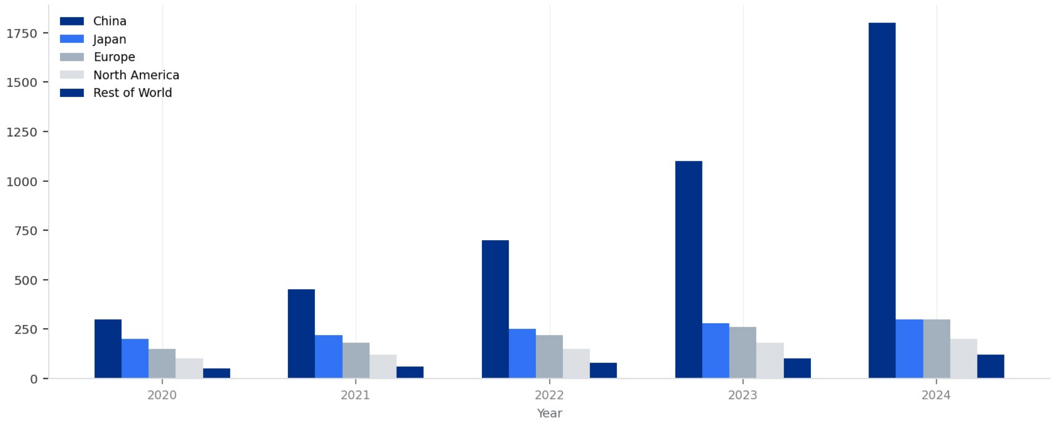

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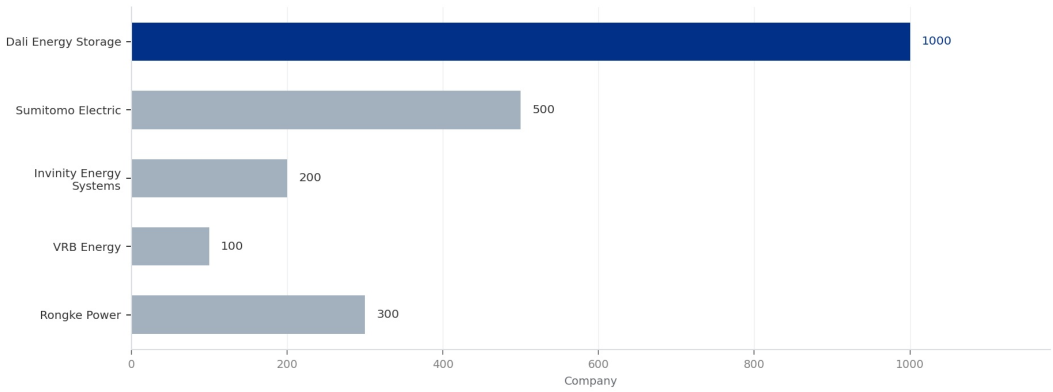

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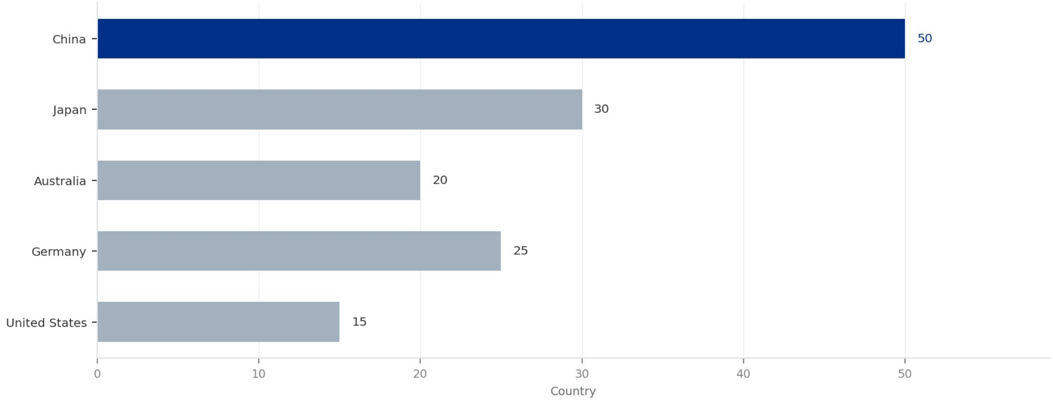

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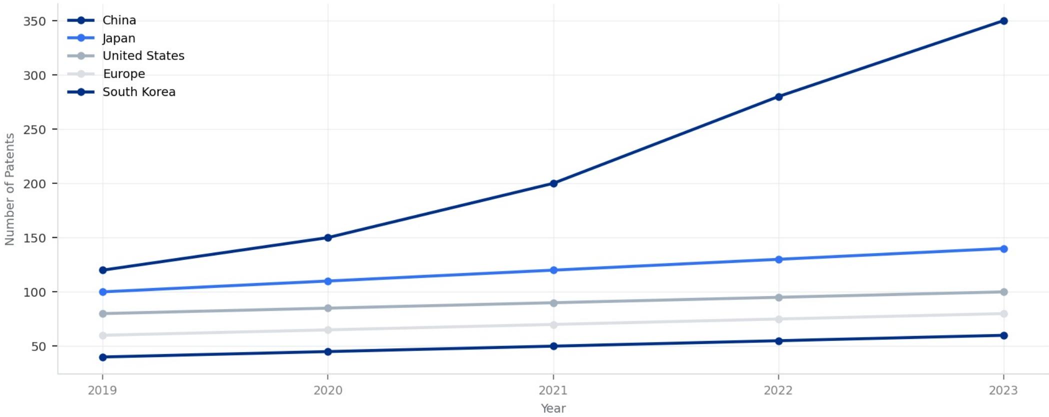

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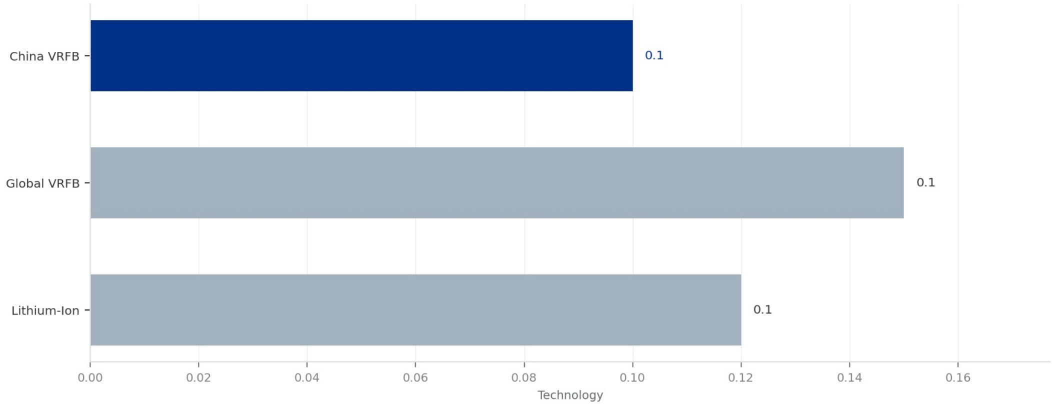

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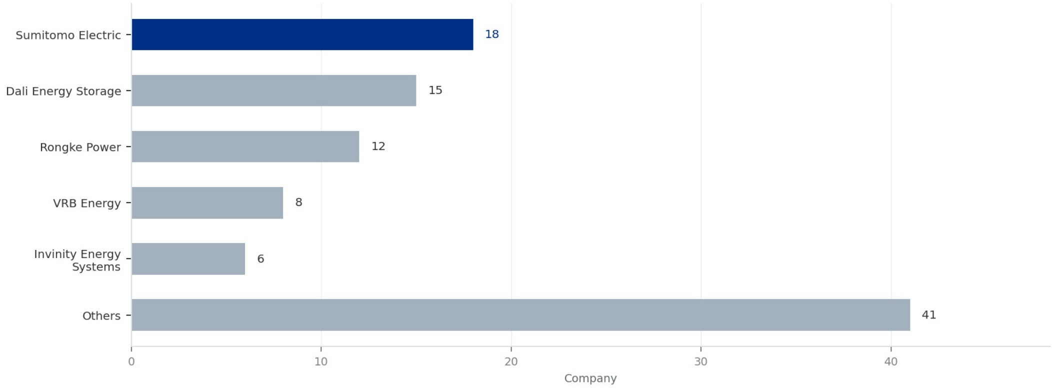

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